CSD416-Project Phase II

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MANGALAM COLLEGE OF ENGINEERING, ETTUMANOOR

(Affiliated with APJ Abdul Kalam Technological University)

NAVIA: A Voice-Assisted Campus Guide Robot

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CONTENTS

OBJECTIVES

- Design a semi-autonomous campus guide robot that follows a predefined circular path to assist visitors in navigating key locations.
- Enable dual-mode interaction through voice commands and touchscreen input, ensuring accessibility for diverse user preferences.
- Provide dynamic 2D route maps that can be displayed on-screen.
- Utilize Raspberry Pi and Arduino for modular control of motor functions, system coordination, and interface management.

ABSTRACT

- NAVIA is a semi-autonomous robot designed to assist campus visitors in locating departments, classrooms, and facilities through voice and touchscreen interaction.
- The robot follows a predefined circular path within a limited radius, ensuring consistent availability in central campus zones.
- Users can engage via touchscreen (Android tablet interface) or voice command (microphone + speech recognition APIs).
- The system uses a Python backend to query a structured college database and deliver real-time responses.

INTRODUCTION

- Helping campus visitors easily find departments, rooms, and facilities.
- The robot moves in a circular path to stay visible and accessible.
- Users can ask questions using voice or touchscreen.
- Answers are given using a college database and Python logic.
- Uses Raspberry Pi and Arduino to control movement and system functions.
- Shows 2D route maps on a tablet.
- Supports SDG goals: Education, Innovation, and Smart Cities.

PROBLEM DEFINITION

- Campuses are large and difficult for newcomers to navigate.
- Traditional maps and signboards are not interactive and may fail to provide real-time or personalized assistance.
- Existing methods are not accessible for differently abled individuals.
- Existing systems lack both touchscreen and voice interaction together.
- Need for a low-cost and user-friendly navigation robot suitable for real-world campus use.

LITERATURE REVIEW

Sl No	TITLE	AUTHOR	KEY CONCEPT	RESEARCH CONTRIBUTION
1	How Adaptation and User Engagement Affect Trust in Audio Guide Agents [2025]	Mari Saito, Seiji Yamada	Audio guide agents are trusted more when they adapt to the environment and are easy for users to interact with.	Methodology contributed is to how adaptation and user engagement influence user trust in audio guide agents.
2	System and User Strategies to Repair Conversational Breakdowns of Spoken Dialogue Systems: A Scoping Review[2024]	E. Alghamdi, M. Halvey, E. Nicol	Shows the spoken dialogue systems and users fix communication errors to continue smooth conversations.	Methodology contributed is to systematic literature search, screening, data extraction, and thematic classification of system and user repair strategies in spoken dialogue systems.

LITERATURE REVIEW

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3	The Impact of Perceived Tone, Age, and Gender on Voice Assistant Persuasiveness [2024]	S. B. H. Pias, R. Huang, M. Kim, A. Kapadia	Users trust intelligent systems more when the system is transparent, protects privacy, handles errors well, and is designed around human needs.	Methodology contributed is to user-centered experiments and analysis of user interactions to examine how trust, transparency, and privacy affect human interaction with intelligent systems.
4	Developing Trustworthy Artificial Intelligence: Insights from Human-AI Trust Research [2024]	Y. Li, B. Wu, Y. Huang, S. Luan	Trustworthy AI is built by user characteristics and AI system qualities, the context in which humans interact with AI.	Methodology contributed is to analyze the existing human-AI trust studies and derive a framework for designing trustworthy AI systems.

EXISTING SYSTEM VS PROPOSED SYSTEM

Existing System

- Static campus maps lack interactivity and real-time guidance.
- Limited user engagement.
- No adaptive audio or voice interaction in current campus navigation tools
- Limited support for visually impaired or first-time visitors.

Proposed system

- Users can ask questions using a voice or touchscreen interface.
- It also supports touch screen–based interaction via a mobile application.
- Uses Arduino Uno as the main control unit.
- Ultrasonic sensors provide obstacle detection and safety.
- Designed to be low-cost, reproducible, and aligned with SDG goals

SYSTEM SPECIFICATION

Hardware

- Raspberry pi 5 model 8 GB RAM
- USB-C PD Power Supply for Raspberry Pi 5 (Black, IN plug)
- L298N Dual H-Bridge DC/Stepper Motor Driver Module
- 12V 100 RPM Johnson Geared DC Motor (Encoder compatible) – 2 Nos
- Arduino mega
- 7-inch LCD screen display with HDMI
- 85mm Robot Smart Car Wheel
- 6mm Hex Motor Coupling for Smart Car Wheel
- Official Raspberry Pi 5 Active Cooler
- 24V/12V to 5V 5A DC-DC Power Converter
- Microphone
- Ultrasonic sensors
- Lithium battery

SYSTEM SPECIFICATION

Software

- Windows
- Raspberry Pi OS (64-bit)
- Arduino IDE
- Speech Recognition APIs
- Tkinter

WORKING PRINCIPLE

Central Processing Unit

- The Raspberry Pi 5 (8GB RAM) acts as the main processing unit of the system.
- It handles user interaction, speech-to-text processing, database lookup, and display control.

User Interaction Mechanism

- A 7-inch HDMI touchscreen display is used to accept touch input from users and to display campus maps, directions, and location information.
- For voice interaction, the user's speech is captured and converted into text using a speech-to-text service, enabling hands-free operation.

Query Processing (Rule-Based)

- The received input (voice or touch) is processed using **rule-based keyword matching**.
- The keywords are compared with entries stored in a **predefined campus database**. Once a match is found, corresponding navigation details are retrieved.

Guidance Output

- The system provides audio guidance through the Raspberry Pi and visual guidance using directional maps and text displayed on the touchscreen.
- This dual-mode output improves clarity and user trust.

Motion Control System

- An Arduino Mega 2560 is used exclusively for controlling hardware movement.
- The L298N motor driver controls the direction and speed of DC motors.
- Johnson geared DC motors drive the wheels for robot movement.
- Robot wheels, couplers, and brackets provide mechanical stability.

Predefined Navigation Path

- The robot moves along a fixed circular path inside the campus.
- This ensures predictable movement, easy maintenance, and continuous availability without complex path planning.

Obstacle Detection and Safety

- An **ultrasonic sensor** continuously monitors the front area.
- If an obstacle is detected, the Arduino temporarily stops or adjusts movement, ensuring safe operation in crowded environments.

Servo-Based Interaction (Optional)

- MG995 servo motors can be used for simple mechanical actions such as hand gestures or pointing mechanisms to enhance user interaction.

System Continuity

- After completing user interaction, the robot resumes its predefined route, allowing it to assist multiple users sequentially without manual intervention.

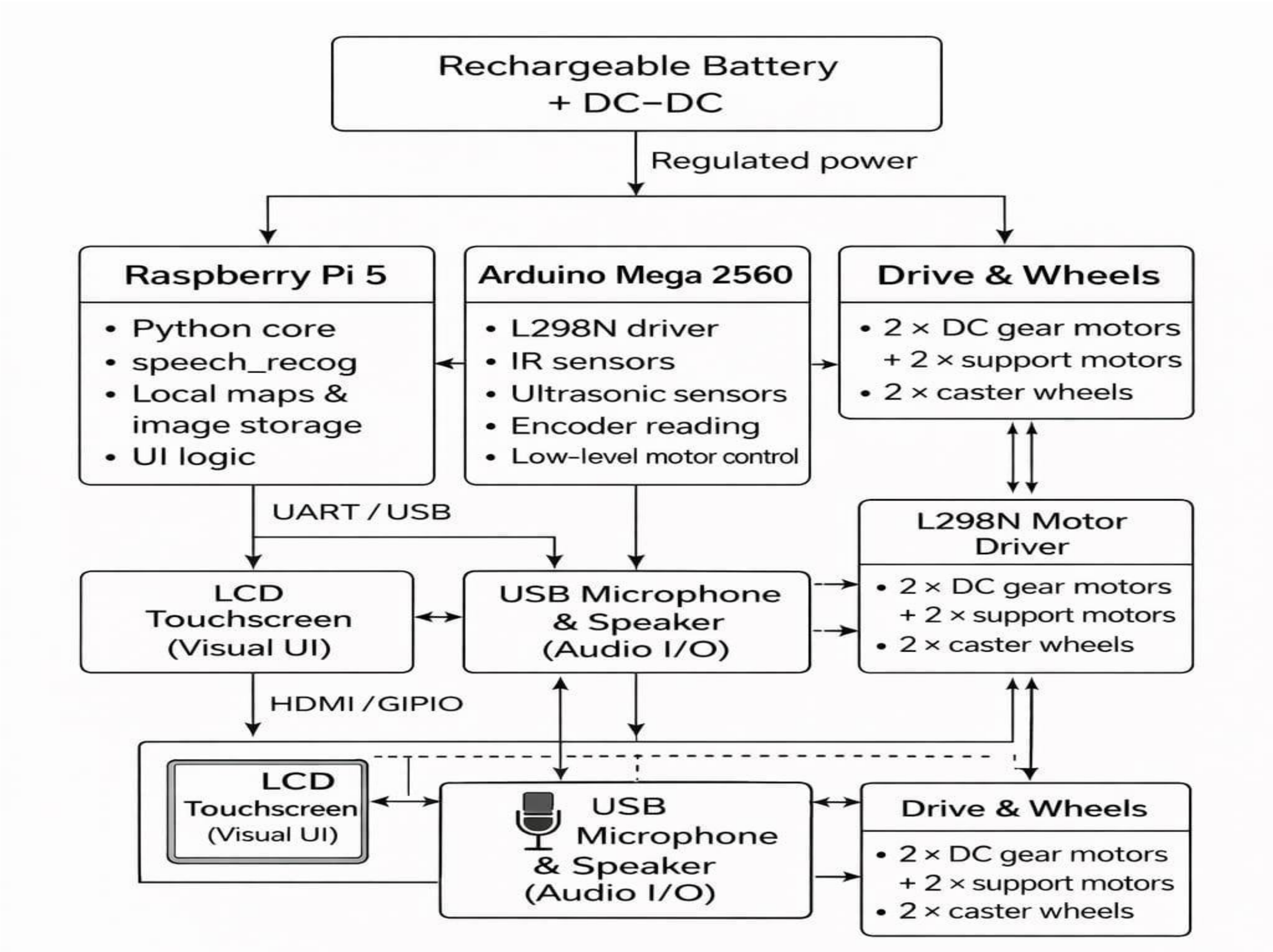
Power Supply and Battery Management

- The system uses a separate battery unit as the primary power source.
- DC-DC buck converters (LM2596 / XY-3606) regulate voltage levels.
- 5V supply for Raspberry Pi and display.
- Required voltage for Arduino, motors, and sensors
- This ensures stable power delivery, protects components, and improves battery efficiency.

Cooling and Reliability

- An active cooler for Raspberry Pi 5 prevents overheating during continuous operation.
- Improving system reliability and lifespan.

SYSTEM ARCHITECTURE



DEVELOPMENT TOOLS

BACK-END:

- VS code
- API
- Tkinter
- python
- Arduino IDE

MODULES

The proposed method uses the following modules:

1. User Interaction Module
2. Query Processing Module
3. Campus Information Database Module
4. Movement Module
5. Response Delivery Module
6. Integration and Control Module

User Interaction Module

- The module enables the users to communicate with the campus guide robot in a natural and safe manner.
- The system allows the user to interact with the system through the use of voice and touchscreen.
- The module they have then interprets the commands, standardizes the form of the message, and forwards it to the back-end system.

Query Processing Module

- Responsible for knowing precisely what the users require and providing the appropriate responses.
- The entire backend is basically Python-based to execute the tasks that handle, e.g., query understanding and access the information stored in the campus database.
- Searching for the appropriate answer in a database and providing the screen or audio system of the robot.

Campus Information Database Module

- Detailed information on classrooms and libraries formatted in a well-defined and easily retrievable manner.
- The database is closely linked to the Que Query Processing and the AI Module.
- The module assists the robot to provide the intelligent responses and aids the visitor to navigate the camp efficiently.

Movement Module

- Used to control the physical motion of the campus guide robot.
- Arduino and Raspberry Pi control boards to command the motors.
- Obstacle detectors can also be added to the module to avoid a collision during movement.

Response Delivery Module

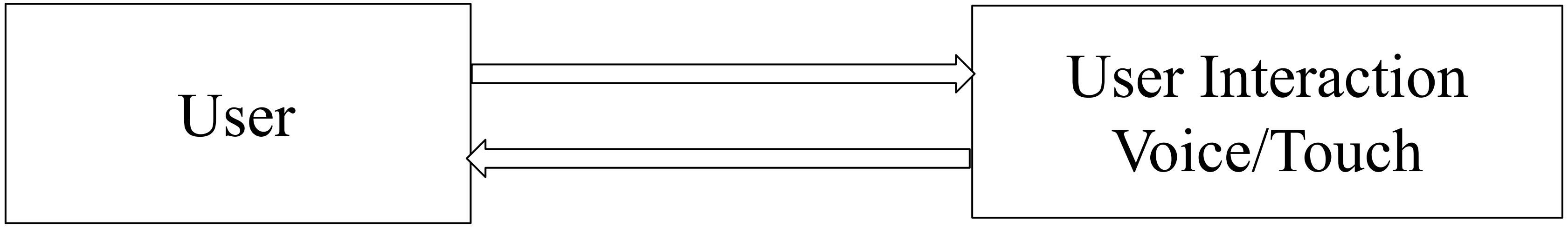
- Provide responses and advice produced by the system back to the user.
- This module will ensure that the user receives clear and understandable instructions on the screen or listens to the instructions in the form of a voice.
- Designed to integrate and monitor the activities of all modules to ensure seamless operation of the system.

Integration and Control Module

- Coordinates all the hardware and software components of the campus guide robot
- Provides smooth communication between the user interface powered by the AI-powered query processing module, the campus database, and the robotic and navigational system.
- The module synchronizes user input, artificial intelligence interpretation, and physical navigation to the effect that the robot accurately responds, moves efficiently, and provides information in a timely manner.
- Integrating all the modules, thus offering an integrated and reliable user experience.

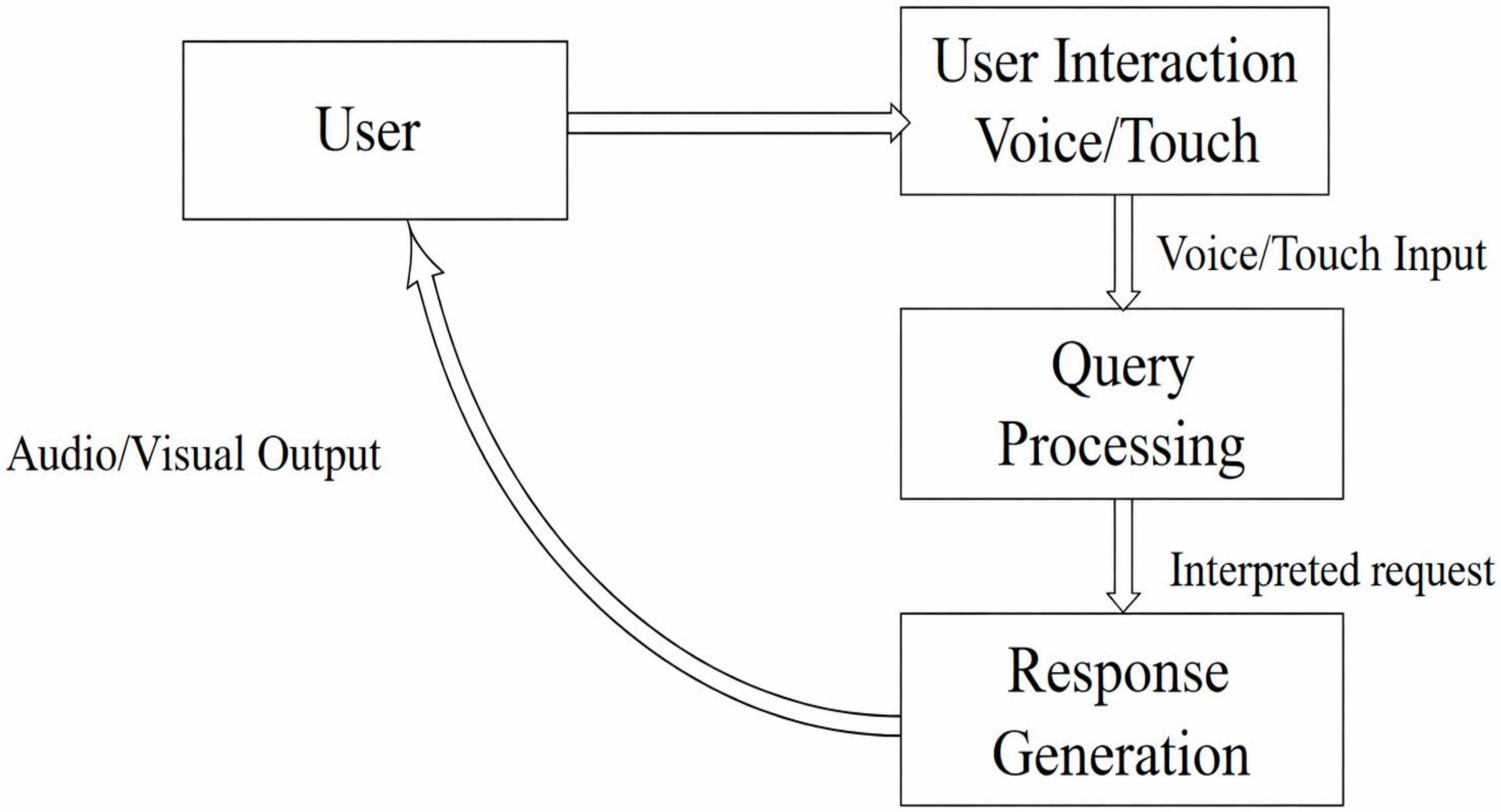
DETAILED DESIGN

DFD LEVEL 0:



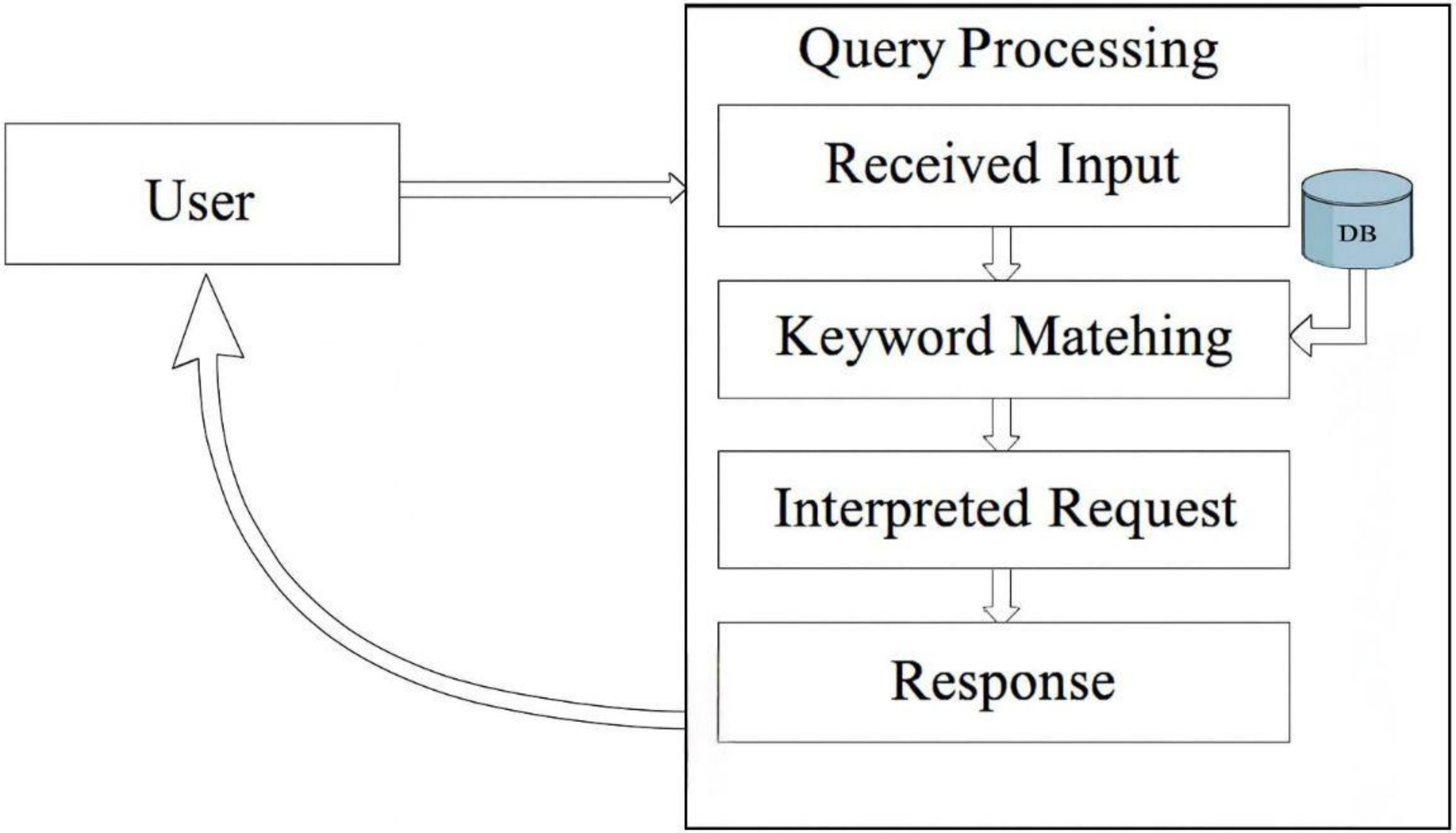
DETAILED DESIGN

DFD LEVEL 1:



DETAILED DESIGN

DFD LEVEL 2:

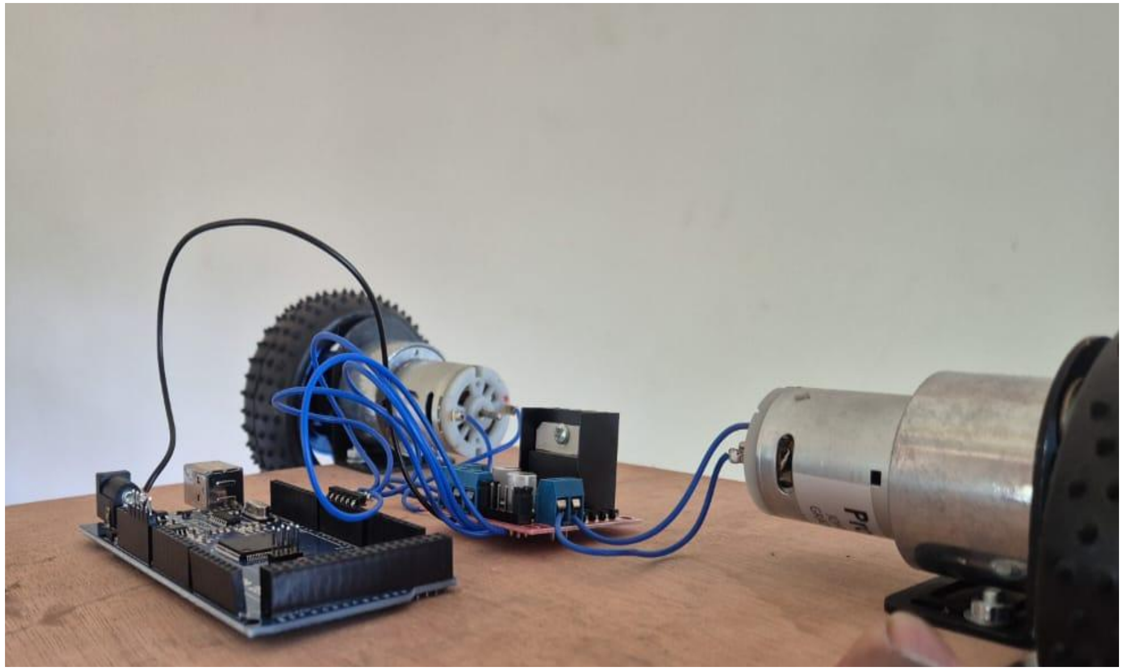


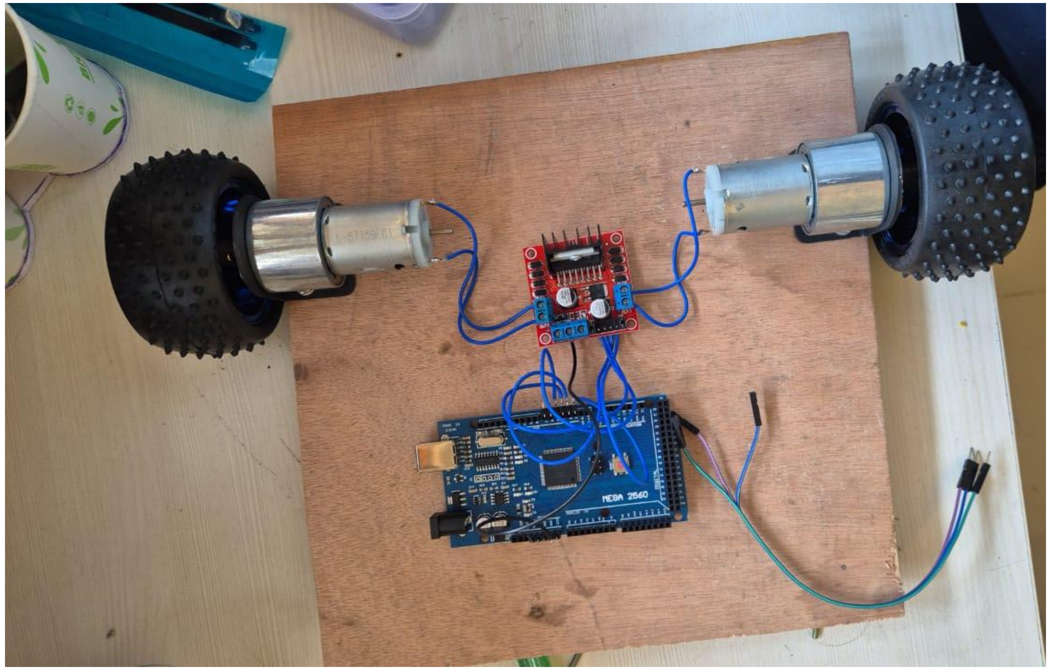
EXPERIMENTS & RESULTS

- Basic robotic mobility platform designed to test the movement control of the proposed campus guide robot.
- Components are mounted on a wooden base to provide mechanical support and reduce vibration during operation.
- An Arduino microcontroller is used as the main control unit to generate control signals for motor operation.
- A motor driver module is interfaced with the Arduino to supply sufficient power to the DC motors and to control their direction and speed.
- Two DC geared motors are fixed on opposite sides of the base and connected to rubber wheels, enabling differential drive motion.
- The motors receive commands from the Arduino through the motor driver, allowing the robot to move forward, turn, and follow.

EXPERIMENTS & RESULTS

- Proper wiring is established between the Arduino, motor driver, and motors to ensure reliable signal transmission.



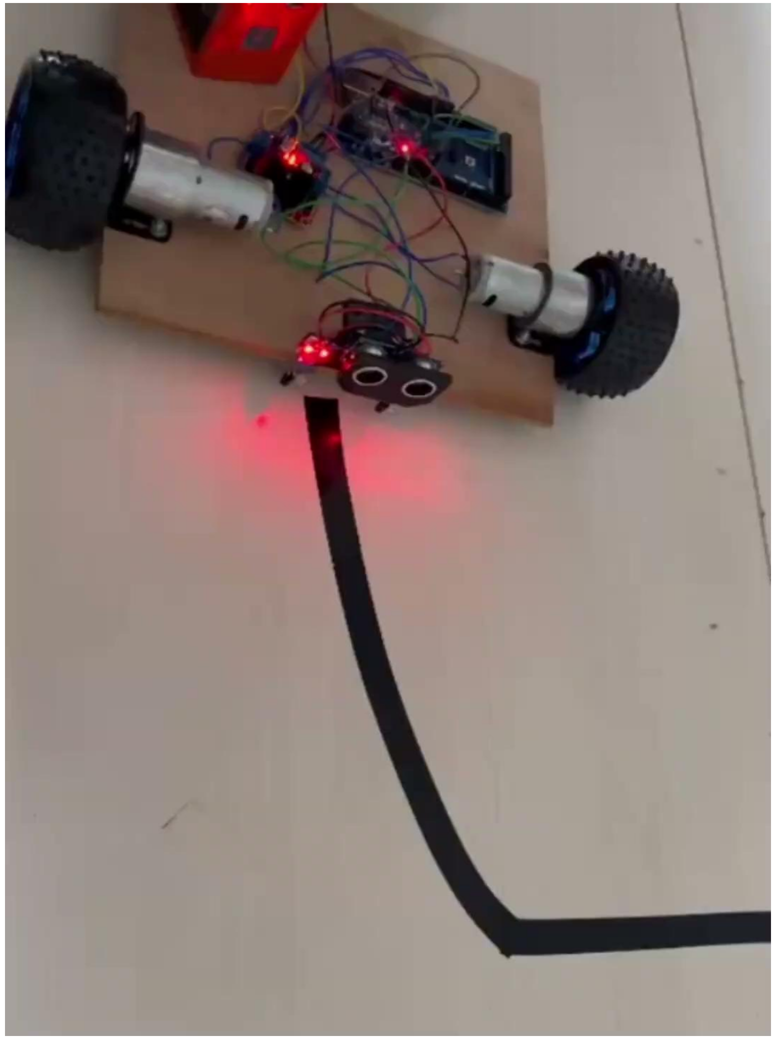


EXPERIMENTS & RESULTS

The robot was successfully designed and implemented as an autonomous line-following robot with obstacle detection capability.

- The robot accurately followed the predefined black line track on a light-colored floor.
- Successfully navigated straight paths, curves, and slight turns without deviation.
- The IR sensor module detected the line consistently and adjusted wheel movement accordingly.
- The ultrasonic sensor detected obstacles placed along the path.
- The robot maintained stable movement under normal indoor lighting conditions.
- The interactive display screen enhanced user interaction by providing a friendly interface.

EXPERIMENTS & RESULTS



CONCLUSION & FUTURE SCOPE

- In conclusion, this project successfully demonstrates how integrating voice recognition with sensor-based navigation can create an intelligent, interactive assistant for campus visitors.
- Delivers a reliable, user-friendly, and low-cost campus guide robot that improves accessibility, automation, and navigation.
- Users can engage with NAVIA through voice or touchscreen input, receiving real-time visual and audio guidance supported by a Python backend and Raspberry Pi for processing.
- Advanced natural language processing can be added to support more natural conversations
- Additional features like facial recognition for personalized guidance, mobile app integration, and wheelchair-accessible route suggestions can further improve usability.
- Better conversation repair features can be added when the robot does not understand users.

REFERENCES

[1] M. Saito and S. Yamada, "How adaptation and user engagement affect trust in audio guide agents," IEEE Access, vol. 13, pp. 81553–81568, May 2025, doi: 10.1109/access.2025.3567067.

[2] M. Rheu, J. Y. Shin, W. Peng, and J. Huh-Yoo, ''Systematic review: Trust-building factors and implications for conversational agent design,'' Int. J. Hum.–Comput. Interact., vol. 37, no. 1, pp. 81–96, Jan. 2021.

[3] N. Zimmerman, J. Loo, A. Agrawal, and D. Hsu, "SignLoc: Robust localization using navigation signs and public maps," *arXiv*, Aug. 2025, doi: 10.48550/arXiv.2508.18606.

[4] K. Zhu and T. Zhang, "Deep reinforcement learning based mobile robot navigation: A review," *Tsinghua Sci. Technol.*, vol. 26, no. 5, pp. 674–691, Apr. 2021, doi: 10.26599/TST.2021.9010012.

[5] B. K. Riley and A. Dixon, ''Emotional and cognitive trust in artificial intelligence: A framework for identifying research opportunities,'' Current Opinion Psychol., vol. 58, Aug. 2024, Art. no. 101833.

[6] H. Wang, A. H. Tan, A. Fung, and G. Nejat, "X-Nav: Learning end-to-end cross-embodiment navigation for mobile robots," *CoRR*, vol. abs/2507.14731, 2025, doi: 10.48550/arXiv.2507.14731.

[7] M. Rheu, J. Y. Shin, W. Peng, and J. Huh-Yoo, ''Systematic review: Trust-building factors and implications for conversational agent design,'' Int. J. Hum.–Comput. Interact., vol. 37, no. 1, pp. 81–96, Jan. 2021.

[8] Z. Hu, T. Xu, X. Xiao, and X. Wang, "CARoL: Context-aware adaptation for robot learning," *IEEE Robot. Autom. Lett.*, vol. 10, no. 11, pp. 12063–12070, 2025, doi: 10.1109/LRA.2025.3619779.

[9] A. Kumar, S. Luthra, S. K. Mangla, and Y. Kazancoglulu, "COVID-19 impact on sustainable production and operations management," *Sustain. Oper. Comput.*, vol. 1, pp. 1–7, Jan. 2020.

[10] M. Cardona, A. Palma, and J. Manzanares, "COVID-19 pandemic impact on mobile robotics market," in Proc. IEEE ANDESCON, Oct. 2020, pp. 1–4.

[11] H. Rashidi, F. Matinfar, and F. Parand, “Automated guided vehicles—A review on applications, problem modeling and solutions,” *Int. J. Transp. Eng.*, vol. 8, no. 3, pp. 261–278, 2021.

[12] J.E. Sierra-García and M. Santos, “Mechatronic modelling of industrial AGVs: A complex system architecture,” *Complexity*, vol. 2020, pp. 1–21, Dec. 2020.

[13] X. Chen, F. Yang, and C. Wang, “iA*: Imperative learning-based A* search for path planning,” *IEEE Robot. Autom. Lett.*, vol. 10, no. 12, pp. 12987–12994, 2025, doi: 10.1109/LRA.2025.3625500.

[14] C. Yang, K. Yuan, Q. Zhu, W. Yu, and Z. Li, “Multi-expert learning of adaptive legged locomotion,” *Sci. Robot.*, vol. 5, no. 49, 2020, Art. no. eabb2174.

[15] C. R. d. Cos and D. V. Dimarogonas, “Adaptive cooperative control for human-robot load manipulation,” *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 5623–5630, Apr. 2022.

[16] C. Mavrogiannis, “Core challenges of social robot navigation: A survey,” 2021, *arXiv:2103.05668*.

[17] R. Han, “Reinforcement learned distributed multi-robot navigation with reciprocal velocity obstacle shaped rewards,” *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 5896–5903, Jul. 2022.

[18] Z. Liu, “Visuomotor reinforcement learning for multirobot cooperative navigation,” *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 4, pp. 3234–3245, Oct. 2022

[19] S. Lee, H. Yu, G. Kim, and S. Choi, “LAMP: Implicit language map for robot navigation,” *IEEE Robot. Autom. Lett.*, vol. 10, no. 12, pp. 12365–12372, 2025, doi: 10.1109/LRA.2025.3619820.

[20] C. Xu, C. Amato, and L. L. S. Wong, “Robot navigation in unseen environments using coarse maps,” in *Proc. IEEE Int. Conf. Robot. Automat.*, 2024, pp. 2932–2938.

THANK YOU!